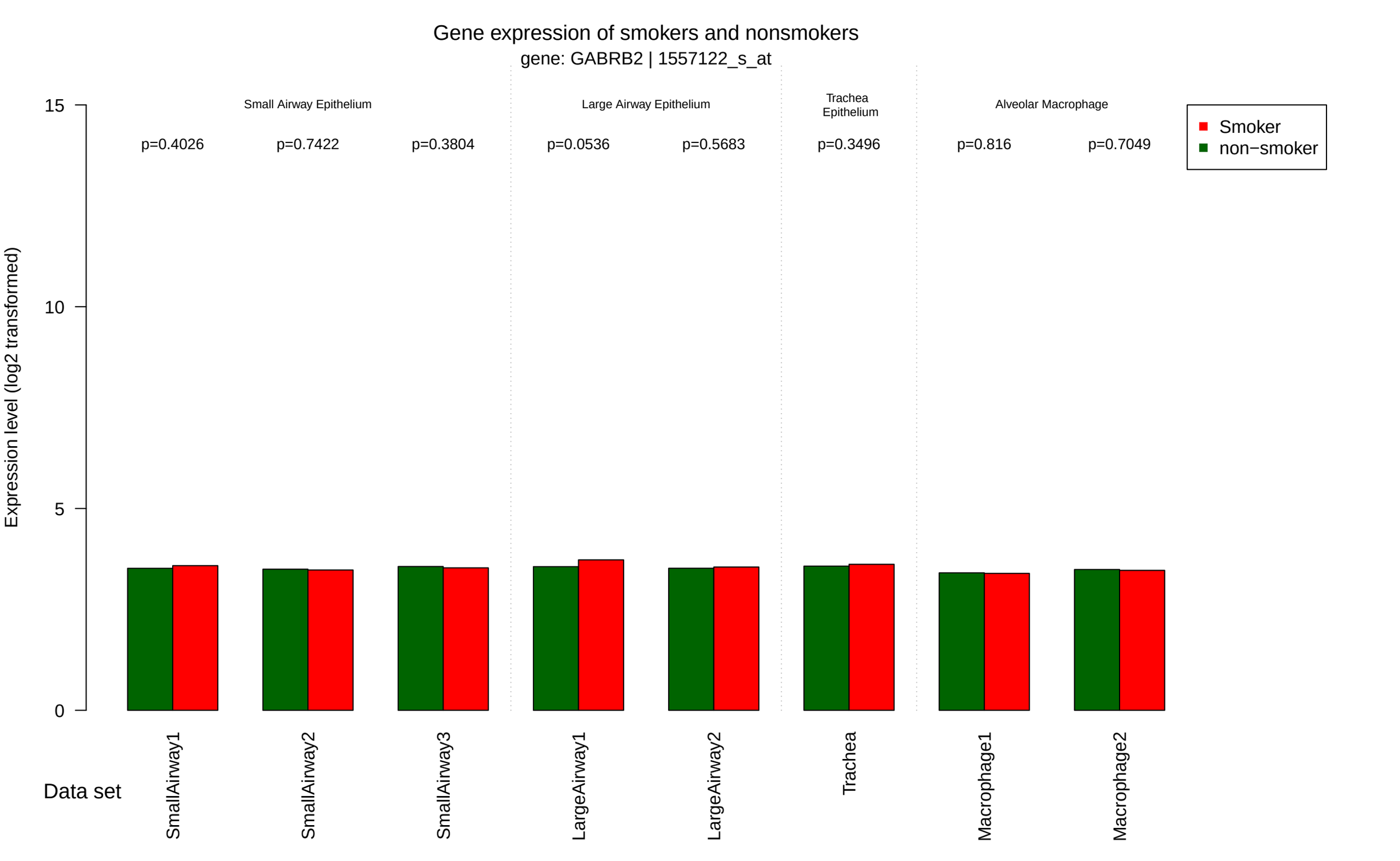
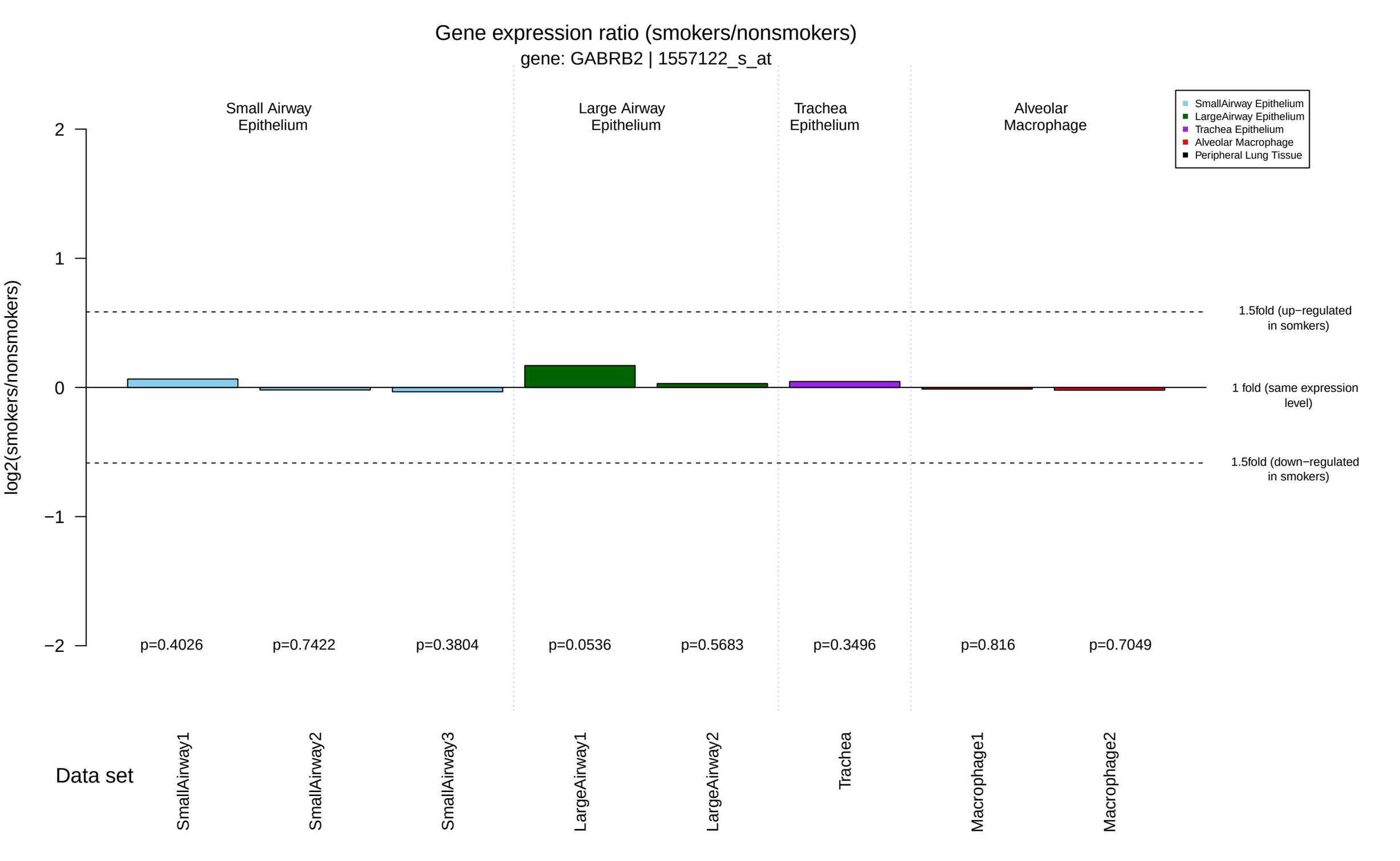
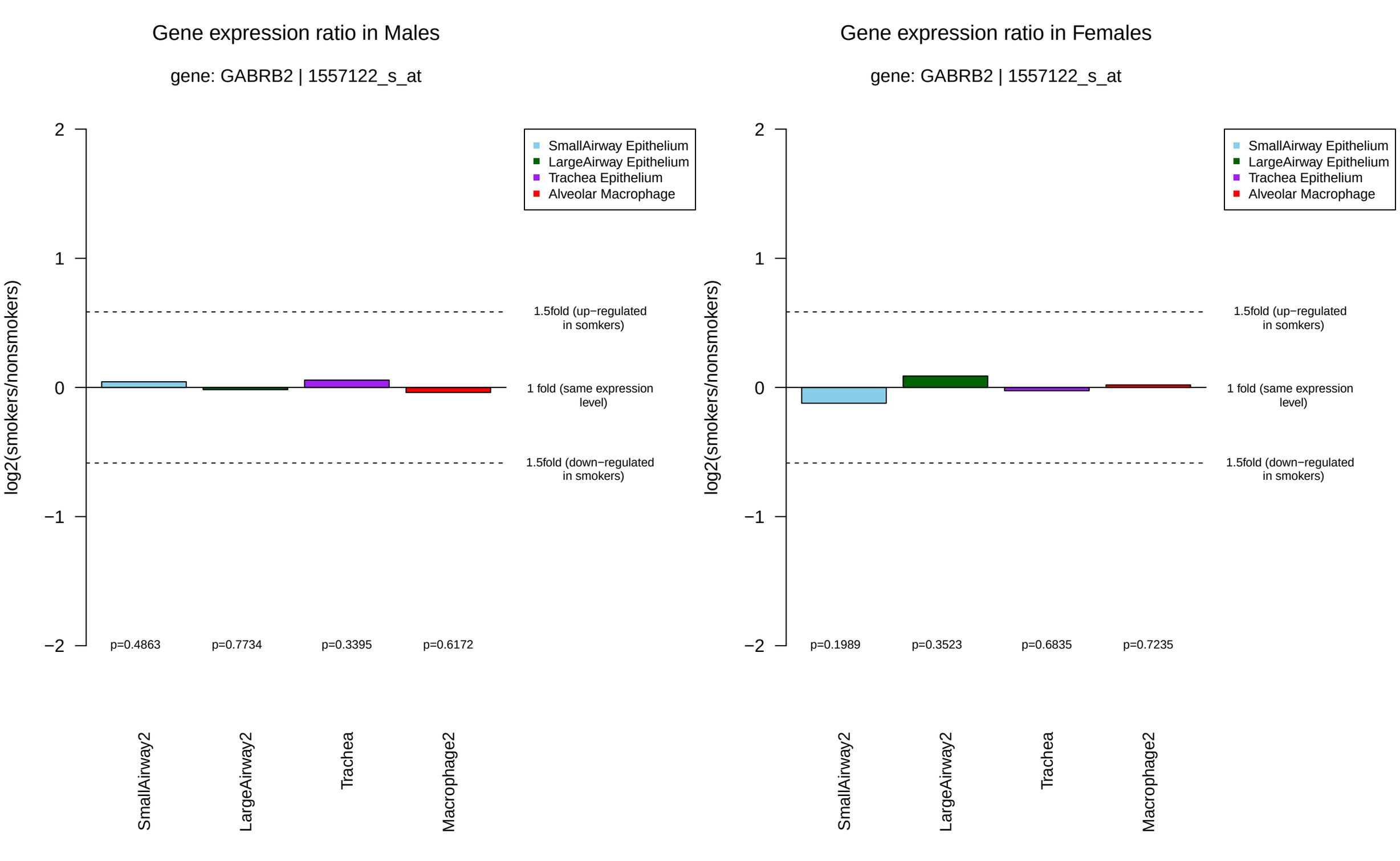
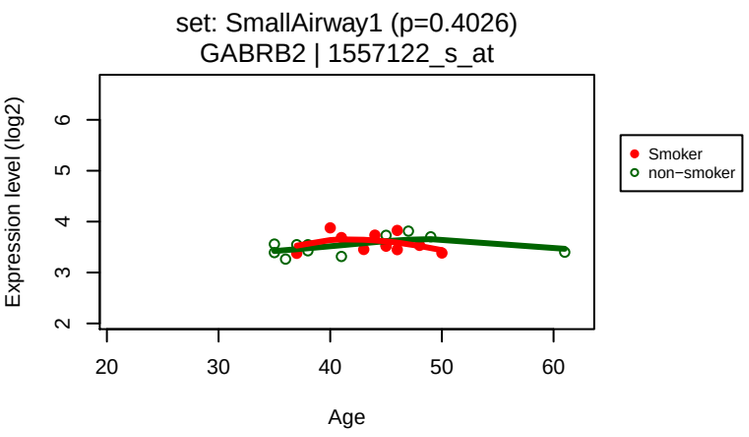
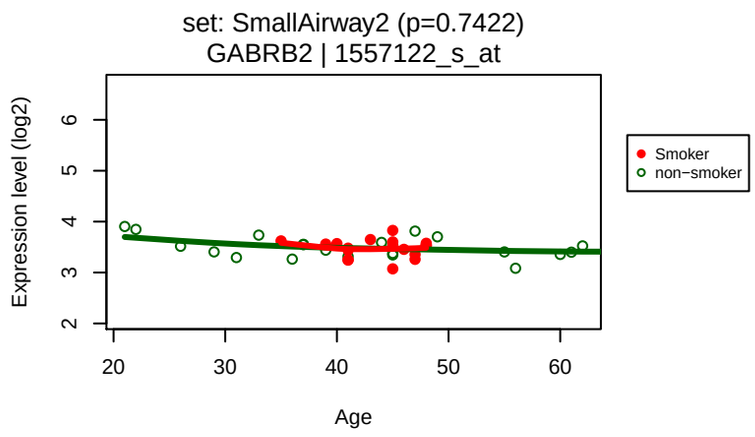
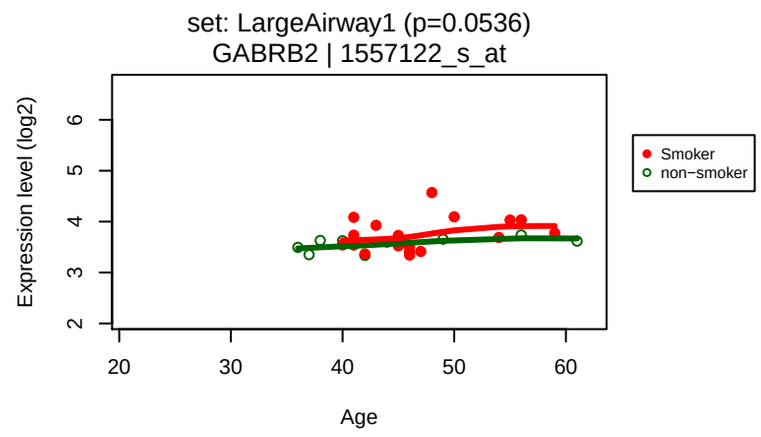

Gene expression of smokers and nonsmokers
gene: GABRB2 | 1557122_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4026
p=0.7422
p=0.3804
p=0.0536
p=0.5683
p=0.3496
p=0.816
p=0.7049
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

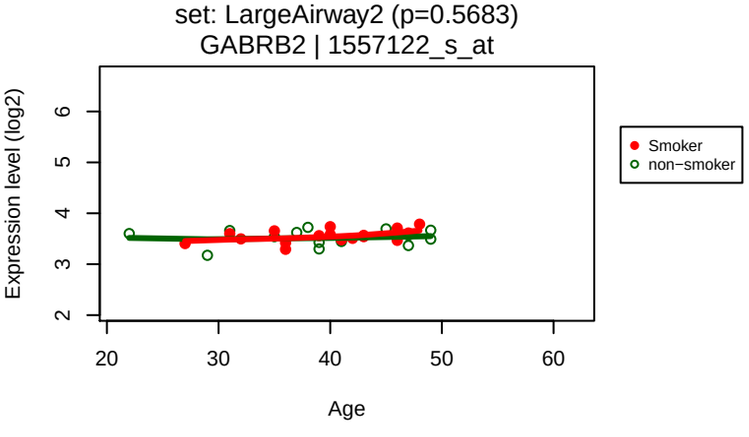

Gene expression ratio (smokers/nonsmokers)
gene: GABRB2 | 1557122_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4026
p=0.7422
p=0.3804
p=0.0536
p=0.5683
p=0.3496
p=0.816
p=0.7049
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

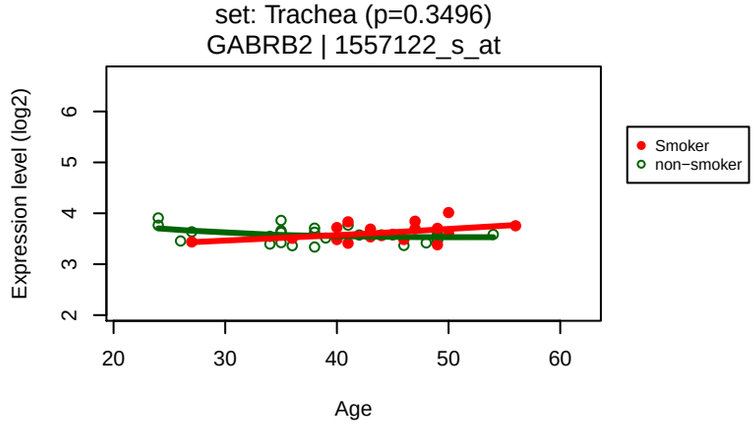

Gene expression ratio in Males
gene: GABRB2 | 1557122_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4863
p=0.7734
p=0.3395
p=0.6172
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GABRB2 | 1557122_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1989
p=0.3523
p=0.6835
p=0.7235
SmallAirway2
LargeAirway2
Trachea
Macrophage2

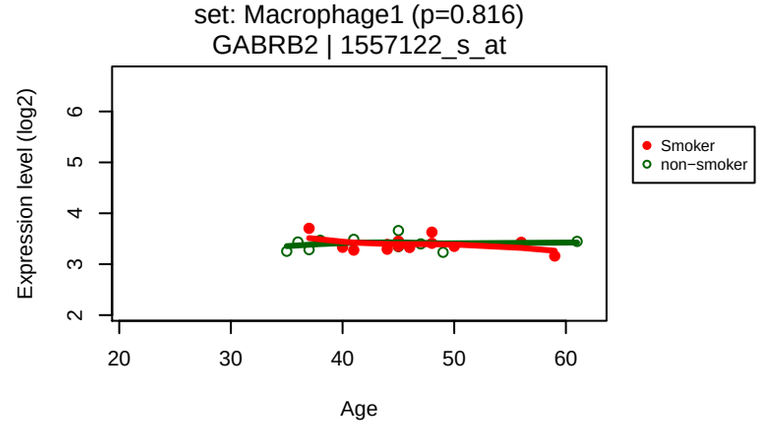

set: SmallAirway1 (p=0.4026)
GABRB2 | 1557122_s_at
Expression level (log2)
Age
Smoker
non−smoker

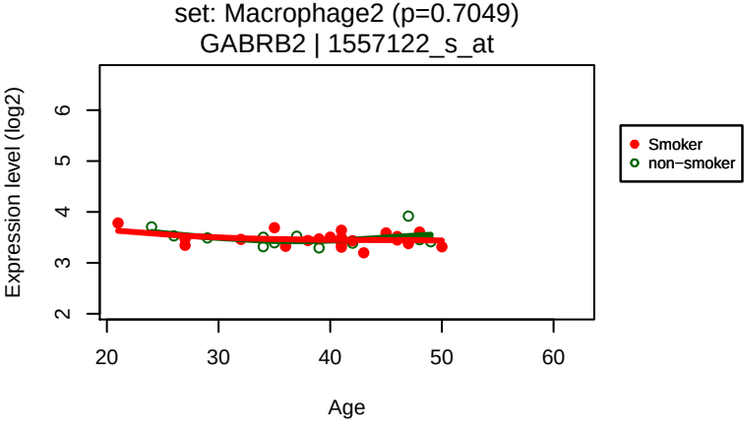

set: SmallAirway2 (p=0.7422)
GABRB2 | 1557122_s_at
Expression level (log2)
Age
Smoker
non−smoker

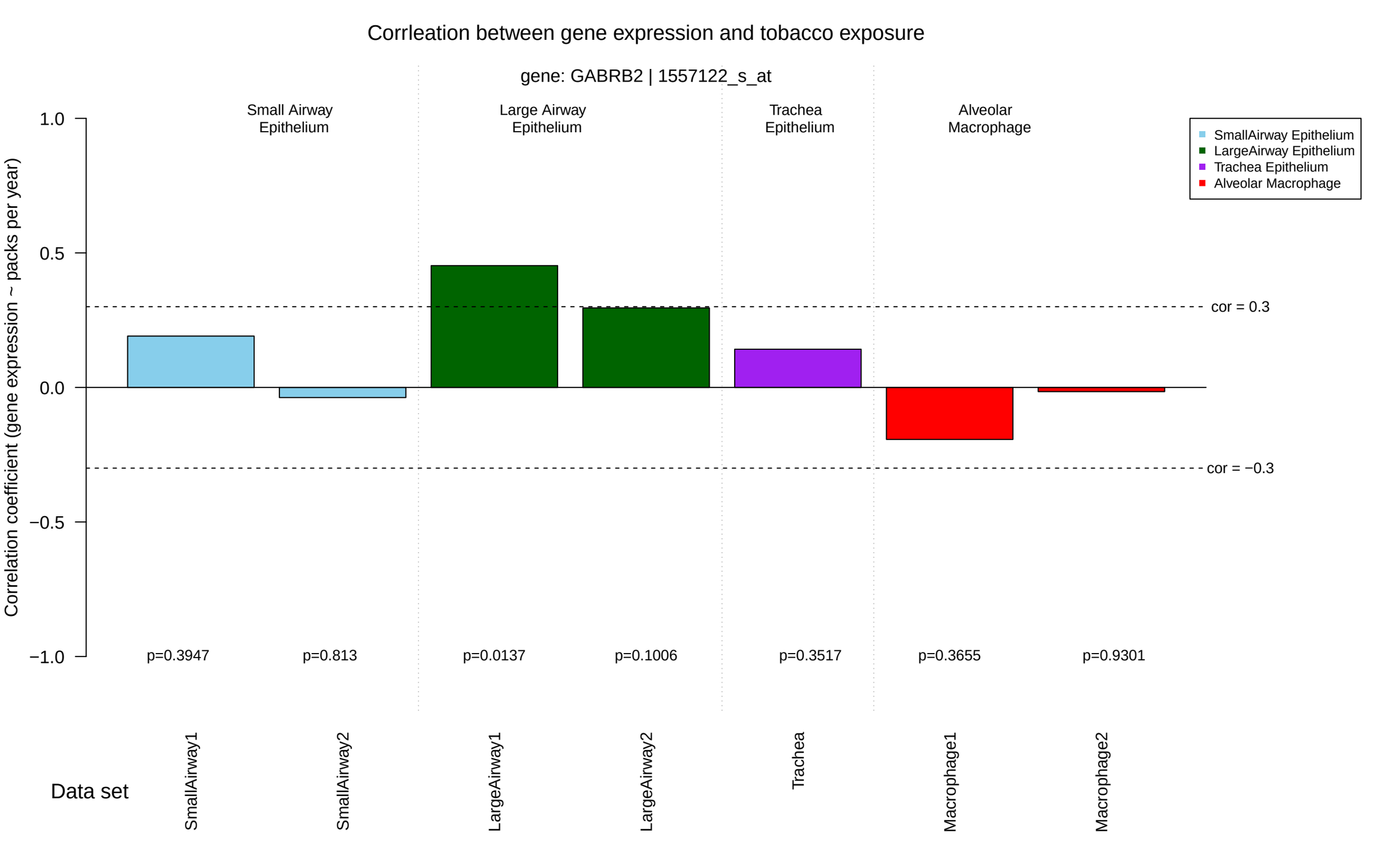

set: LargeAirway1 (p=0.0536)
GABRB2 | 1557122_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5683)
GABRB2 | 1557122_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3496)
GABRB2 | 1557122_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.816)
GABRB2 | 1557122_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7049)
GABRB2 | 1557122_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GABRB2 | 1557122_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3947
p=0.813
p=0.0137
p=0.1006
p=0.3517
p=0.3655
p=0.9301
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2